

Three creative ideas for your graduation capstone project:

## 1. "AI-Sprint" – Agile Project Management & Task Automator

This project is a collaborative tool for development teams that uses AI to transform high-level project ideas into actionable technical tasks.

- **Frontend (React & TypeScript):** Use **TypeScript** for robust, type-safe components. The UI should be built with **Bootstrap** for a mobile-first, responsive grid system and include **Advanced Components** like Modals and Navbars for a professional feel.
- **Prompt Engineering Integration:** Implement **intelligent, context-aware prompts** that take a user's "Project Goal" and use **content generation** to break it down into specific user stories and acceptance criteria.
- **Backend & State Management:** Manage the application state using **Redux Toolkit** to handle real-time task updates across the team. The backend will be a **Node.js and Express** server providing a **RESTful API** to store tasks in a **MongoDB** database.
- **DevOps & Reliability:** The entire application should be **containerized using Docker** to ensure consistent deployment environments.

## 2. "PortfolioGenie" – AI-Powered Developer Portfolio Builder

A platform that helps junior developers create professional, high-performance portfolios by analyzing their GitHub activity and generating optimized content.

- **UX/UI Design:** Follow the **design thinking process** to create user personas and wireframes that prioritize developer aesthetics and accessibility.
- **Core Logic (React & Hooks):** Utilize **React Hooks** (useState, useEffect, and custom hooks) to create a dynamic, multi-step form where users can input their projects. Use **React Router** for seamless programmatic navigation between the builder and the live preview.
- **Prompt Engineering Integration:** Leverage AI to optimize project descriptions for SEO and readability, and use **prompt engineering techniques** to suggest professional "About Me" sections based on the user's coding style.
- **Best Practices:** Adhere to **industry-standard coding conventions** and design patterns to ensure the generated portfolio code is clean and maintainable.

## 3. "DevReview AI" – Collaborative Code Style & Documentation Hub

A centralized hub for teams to upload code snippets and receive AI-driven feedback on code style, patterns, and documentation.

- **Frontend (React & Advanced CSS):** Build a highly interactive interface using **HTML5 Canvas** or **SVG** for visualizing code structure and **CSS Animations/Keyframes** for smooth UI transitions.
- **Backend (Node.js & Security):** Use **Express middleware** to handle file uploads and implement **Authentication/Authorization with JWT** (JSON Web Tokens) to protect private code snippets.
- **Prompt Engineering & Code Quality:** Use AI to analyze uploaded JavaScript or TypeScript code against "Code style, patterns and best practices". The AI will provide suggestions for refactoring and automatically draft **functional documentation** and **unit test stubs** for the user.

- **Testing & Version Control:** Implement **Unit Testing** using **Jest and React Testing Library** to ensure the core logic is error-free. Manage all project iterations through **GitHub** using version control best practices.

## Capstone Integration Checklist

To meet the syllabus requirements for a high-quality project, ensure you:

- Use **TypeScript** to enhance coding efficiency and maintainability.
- Develop **RESTful APIs** with Node.js and Express to communicate with your React frontend.
- Incorporate **Docker Compose** for managing multi-container application setups (e.g., frontend, backend, and database).
- Create comprehensive **Functional Documentation** including user stories and acceptance criteria.